

Denoising Diffusion Probabilistic Models (DDPM):

High Level Summary :

=> Goal : Generate realistic Synthetic data
(Eg : images) from noise.

=> core idea : Destroy image structure gradually
by adding Gaussian noise over
T steps (Forward). then train a
U-Net to learn how to undo Each
noise addition step-by-step (Reverse)

Process Breakdown :

Forward Process (Diffusion) - Fixed, No Learning

[Clean Image x_0] $\rightarrow$ [Add Noise Step 1] $\rightarrow$
... $\rightarrow$ [Pure Gaussian Noise x_T]

Reverse Process (Generative) - Neural Network learns

[Reconstructed x_0] $\leftarrow$ [Remove Noise Step 1] $\leftarrow$
... $\leftarrow$ [Pure Gaussian Noise x_T]

A. Forward Process $q(x_t | x_{t-1})$

Type: Fixed Markov chain (no trainable parameters)

Operation: Gradually adds Gaussian noise governed
by a Variance Schedule $\beta_t \in (0, 1)$
over T steps (typically $T = 1000$).

Single Step Transition:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I})$$

Closed-form shortcut (Direct Sampling at Step t):

Instead of iteratively applying noise t times,
sample x_t directly from x_0 :

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, 1)$$

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s \quad (\text{Cumulative Signal Scale})$$

$\sqrt{\bar{\alpha}_t} \beta x_0$: Remaining signal fraction ($\rightarrow 0$ as $t \rightarrow T$).

$\sqrt{1 - \bar{\alpha}_t} \epsilon$: Cumulative noise fraction ($\rightarrow 1$ as $t \rightarrow T$).

B. Reverse Process $p_\theta(x_{t-1} | x_t)$

Type : Learned Markov chain parameterized by a neural network θ (U-Net).

Goal : Estimate the distribution transition $P_\theta(x_{t-1} | x_t)$ to denoise the sample.

Formulation :

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

Key Insight (Ho et al., 2020) : predicting the added noise ϵ directly gives significantly better

Sample quality than predicting the mean image μ_0 .

Training & Objective Function

Objective Function (L_{simple})

The simplified Variational Lower Bound (VLB) objective is a standard Mean Squared Error (MSE) between the true added Gaussian noise ϵ and the noise predicted by the network ϵ_0 :

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon - \epsilon_0(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2 \right]$$

Training Step Algorithm

1. Sample a Clean Sample $x_0 \sim q(x_0)$ and random timestep $t \sim \text{uniform}(\{1, \dots, T\})$.
2. Sample random Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$.
3. Construct noisy image: $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$.
4. Compute gradient step on $\nabla_{\theta} \|\epsilon - \epsilon_0(x_t, t)\|^2$ to optimize U-Net parameters.

U-Net Architecture & DDPM Adaptation :-

1. Overview

⇒ Original purpose (Ronneberger et al. 2015) :
Fully convolutional network built for biomedical image segmentation (image-to-image dense mapping).

⇒ Role in Diffusion Models (DDPM) : Acts as the backbone neural network $\epsilon_\theta(x_t, t)$ that predicts the noise added at timestep t .

2. Core Architecture Layout

Contracting Path
(Encoder)

Expanding Path
(Decoder)

[High-Res Features] — skip connection → [concatenate & conv]

↓

[Max Pool]

↑
[up-conv]

↓

[Low Res Features] — skip connection → [concatenate & conv]

↓

[Bottleneck]

↑

Key components :

i) Contracting path (Encoder) :

=> operation : Repeated 3×3 conv + ReLU $\rightarrow$
 2×2 Max pooling (stride 2).

=> Effect : Spatial dimensions decrease by half
($H, W \rightarrow H/2, W/2$), channel depth
doubles ($C \rightarrow 2C$). captures high-level
semantic context ("what").

ii) Bottleneck :

=> lowest spatial resolution, highest channel
capacity. Houses the most compressed latent
representation.

iii) Expanding Path (Decoder) :

=> operation : 2×2 Transposed convolution (
or Bilinear upsampling) $\rightarrow 3 \times 3$ conv
+ ReLU.

=> Effect : Spatial dimensions double ($H/2, W/2$
 $\rightarrow H, W$), channel depth halves ($2C \rightarrow C$).

$2c \rightarrow c$). Recovers spatial localization ("where").

iv) Skip connections :

$\Rightarrow$ Copies feature maps directly from encoder layers to corresponding decoder layers via concatenation.

$\Rightarrow$ purpose: preserves fine-grained spatial and edge details lost during downsampling.

3. Adaptation : U-Net in DDPM

In DDPM, the U-Net does not perform pixel classification; instead, it estimates noise:

$$\epsilon_{\theta}(x_t, t) \approx \epsilon.$$

Major Architecture upgrades for DDPM :

[Sinusoidal Embedding + MLP]

Input : Noisy Image (x_t)
Input : Time step (t) $\rightarrow$ [ResNet Block + GroupNorm]



Noise prediction (ϵ) $\leftarrow$ [Self / cross-Attention]

i) Timestep Conditioning (t):

$\Rightarrow$ Timestep t is encoded via Sinusoidal position Embeddings (similar to Transformers) and passed through a Small MLP (Multi-Layer Perception)

$\Rightarrow$ Embeddings are added / projected into every ResNet block across all layers of the U-Net so the model knows the exact noise level to remove.

ii) ResNet Blocks & Group Normalization:

$\Rightarrow$ Standard conv-ReLU blocks are replaced with Residual Blocks (ResNet).

$\Rightarrow$ Uses Group Normalization instead of Batch Normalization to maintain stability across small batch sizes.

iii) Self-Attention Mechanisms:

$\Rightarrow$ Spatial Self-Attention layers are added at lower resolutions (Eg: 16×16 or 8×8).

$\Rightarrow$ Allows the network to model global context and long-range dependencies across the image space.